Does Regulatory Uncertainty Reduce Housing Supply? Evidence from Chicago's Aldermanic Privilege

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Motivation

- Housing prices and rents are skyrocketing in many U.S. cities, including Chicago
  - Limited housing supply in high demand areas is widely accepted as the key driver of these price increases
  - Land use regulations and local control are often blamed for limiting supply
- Most research focuses on formal institutions such as by-right zoning laws or planning commissions
  - But regulatory uncertainty, permitting delays, and discretion are also key determinants of housing supply and are less studied
- **Chicago**: Unique informal institution of *aldermanic privilege*
  - Offers a unique setting where de facto regulatory environment changes discretely at ward boundaries

What is Aldermanic Privilege?

- **Not written in any law** — Chicago has no city charter
- An informal but extraordinarily powerful norm
- Core feature: **Deference**
  - Other aldermen essentially never override a colleague's decision in their ward
  - "Very few things happen that aldermen don't want to happen." (Zhang, 2011)
  - Encyclopedia of Chicago: "little mayors" ruling wards as "fiefdoms"
- **Chicago gives more power to local politicians than many other U.S. cities**:
  - Houston has no explicit zoning code and centralized development approvals
  - Minneapolis has reduced councilmember discretion in recent years
  - Both cities have seen below-median rent and home price growth compared to peer cities (Hembre et al., 2025; Heckler, 2026)

Aldermanic Privilege is a Hotly Contested Issue in Chicago

CITY HALL NEWS ILLINOIS

Chicago wrongly limited affordable housing with aldermanic prerogative, HUD says

Feds say giving council members "a local veto over proposals to build affordable housing" has meant it's "rarely, if ever, constructed in the majority-white wards that have the least affordable housing."

By Brett Chase | Updated Nov 28, 2023, 5:54pm CDT

May 20, 2019

Mayor Lightfoot Signs First Executive Order to Limit Aldermanic Prerogative

In Her First Official Act as Mayor, Lightfoot Instructs Departments to Begin Process of Ending Aldermanic Prerogative as Part of Broader Ethics Reform

Mayor's Press Office 312.744.3334

Download this Press Release

POLITICS

Developer Moves Forward With Lincoln Park Apartment Complex, Setting Stage for Fight Over Aldermanic Prerogative

Heather Cherone | January 9, 2025, 5:00 am

Controversial Old Town Project With Proposed 44-Story Tower Rejected By Alderman

Ald. Brian Hopkins said he won't support Fern Hill's plan for North and LaSalle. But a new design could be coming.

By Patrick Filbin January 24, 2025 Updated January 27, 2025

Sterling Bay's Marcey Street Development Still Stalled After Heated City Council Exchange

The development would include more than 600 apartments in two towers in Lincoln Park near the Chicago River. Ald. Scott Waguespack opposes the plan in its current form.

By Quinn Myers December 11, 2024

This Paper

Research Question: Does the heterogeneity in regulatory uncertainty induced by aldermanic privilege reduce housing supply and raise prices?

Empirical strategy: Segment-localized border design at ward boundaries

- Main density specification uses zoning-code + boundary-segment + year fixed effects (500 ft multifamily sample)

Preview of Results

1. More uncertain aldermen $\Rightarrow$ lower multifamily FAR and DUPAC near ward borders (about 7–10% in border FE)
2. $\sim$2% higher rents on the more stringent side of the border
3. Effect appears to operate through reduced supply

Related Literature

Time-to-Build, Frictions, & Supply Elasticity

- Development lags distort investment and misallocate resources (Budish et al., 2015; Hsieh and Moretti, 2019)
- Permitting lags and "process" are primary drivers of inelasticity (Paciorek, 2013; Gyourko et al., 2021; Molloy et al., 2022)

Regulatory Uncertainty & Discretion

- Uncertainty increases the "option value of waiting" (Cunningham, 2006; Bloom, 2009)
- Discretionary review and local actors amplify delays (Fischel, 2009; Ortalo-Magné and Prat, 2014; Einstein et al., 2019; Khan, 2021; Duranton and Puga, 2023)

Border Discontinuity Designs

- School valuation (Black, 1999; Bayer et al., 2007)
- Spillovers and supply constraints (Turner et al., 2014; Diamond et al., 2019; Kulka et al., 2022)

A "Time-to-Build" Model of Regulatory Uncertainty

Key idea: Regulatory environment $\rightarrow$ effective delay $T(s)$ $\rightarrow$ discounts developer revenue

Developer's problem. A project chooses density $q \geq 0$ with cost $C(q) = F + cq + \frac{\gamma}{2}q^2$.

Revenue arrives upon completion, discounted by regulatory delay:

$$\max_{q \geq 0} \quad e^{-rT(s)}\, p\, q \; - \; C(q)$$

where r = discount rate, $T(s)$ = time-to-build under regulatory environment s

Optimal density (interior solution):

$$q^*(T, p) = \frac{e^{-rT(s)}\, p - c}{\gamma} \qquad \text{if } e^{-rT(s)} p > c, \text{ else } q^* = 0$$

Sufficient statistic: $rT(s)$ summarizes the regulatory wedge — captures delay, redesign, negotiation, financing frictions, and regulatory uncertainty

Testable Predictions

Main comparative static: $\partial q^* / \partial T(s) < 0$, which implies:

1. **Less density** — completed projects have fewer units → Density border-pair FE
2. **Higher rents** — lower aggregate supply $Q = \sum_i q_i^*$, $p'(Q) < 0$ → Rental border-pair FE

Mapping to data:

- $T(s)$ proxied by observed permit-to-completion times
- Regulatory uncertainty index $\approx$ alderman-specific component of $T(s)$
- Cross-boundary variation: different aldermen $\Rightarrow$ discrete changes in $T(s)$

The Challenge: Measuring Regulatory Uncertainty

- Alderman preferences/frictions are latent: need to infer from observed behavior
- **Idea**: Use permit processing times as a **proxy** for regulatory uncertainty and variation in time to build
  - Data: Universe of Permits in Chicago (2006-2025) from Department of Buildings
  - Longer average processing time $\rightarrow$ more back-and-forth, more frictions, more uncertainty
- Focus on **high-discretion permits**: new construction, renovation, demolition

Uncertainty Index Construction

1. **Residualize** processing times on ward characteristics + year × month FEs
   - Controls for baseline differences across wards (income, demographics, etc.)
2. **Estimate alderman fixed effects** on residualized processing times

$$\widehat{\epsilon}_{it} = \mu + \delta_a + \eta_{it}$$

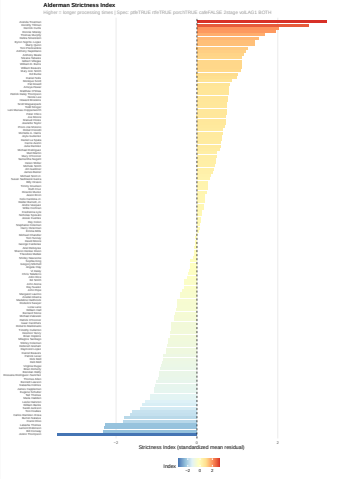
3. **Empirical Bayes shrinkage** to reduce noise (Kline et al., 2022, 2024) then standardize to mean 0, SD 1

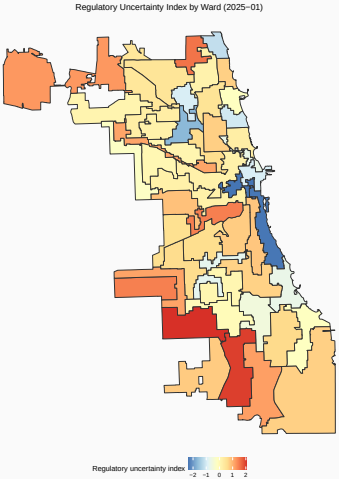
Interpretation: High score = alderman associated with longer processing times

Caveat: This is a *proxy*: likely understates true variation in bureaucratic frictions

- Misses projects that never apply or are discouraged early

Alderman Regulatory Uncertainty Index

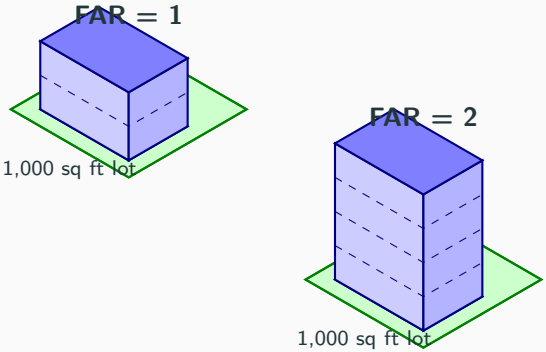




Regression

Validation Checks

Density Outcomes: Floor Area Ratio (FAR)



$$\text{FAR} = \frac{\text{Total Building Floor Area}}{\text{Lot Area}}$$

FAR limits development intensity

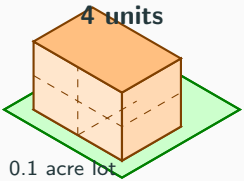
Same lot, different allowed density:

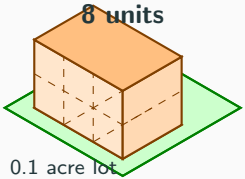
- FAR = 1: Can build 1,000 sq ft
- FAR = 2: Can build 2,000 sq ft

Chicago zoning examples:

Zone	Max FAR	Typical Use
RS-3	0.9	Single-family
RT-4	1.2	Two-flat
RM-5	2.0	Small apartment
RM-6	4.4	Mid-rise

Density Outcomes: Dwelling Units Per Acre (DUPAC)





DUPAC: 40 vs. 80 units/acre (same building volume!)

DUPAC captures population density

$$\text{DUPAC} = \frac{\#\ \text{Dwelling Units}}{\text{Lot Acreage}}$$

Why both measures?

- Same FAR, different DUPAC
- **Developers** care about FAR (total rentable space)
- **Neighbors** care about DUPAC (congestion, parking, schools)

Data: New Multifamily Construction

Source: Cook County Assessor parcel data, 2006-2025

Sample: Multifamily buildings with 2–100 units

- **Single family**: Majority of construction but less variation in density
- **Large developments** (100+ units): Planned Development process with more city oversight

Why multifamily matters:

- Only ∼19% of new construction but a dispraportionate share of total housing supply

Summary statistics:

	All	MF Only
Avg FAR	1.22	2.32
Avg Units	6.65	31.3
Avg Stories	2.27	2.75
Avg DUPAC	28.5	74.5
N	13,236	2,469

Identification Strategy: Border Discontinuity

Idea: Compare new multifamily construction on opposite sides of ward boundaries

- Focus on parcels **within 500 feet** of border
- Main FE stack: **zoning code + boundary segment + construction year**
- Main inference: standard errors clustered by **boundary segment**

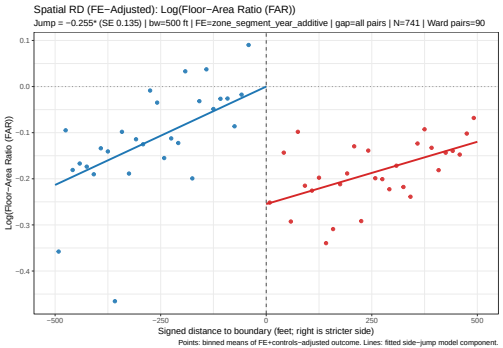
Identifying variation: Within the same boundary segment-year, does the more uncertain side have less dense construction?

Two specifications:

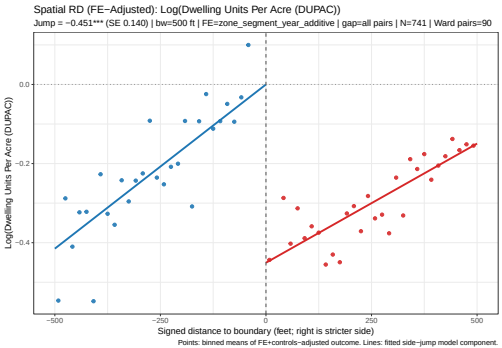
- **Visual RD**: FE-adjusted discontinuity plots under the same FE stack
- **Regressions**: Border FE table with continuous uncertainty index

Visual Evidence: FE-Adjusted Spatial RD at Ward Borders

Log(FAR)



Log(DUPAC)



Notes: Multifamily sample, 500 ft bandwidth, zoning + segment + year FE, segment-clustered SEs.

▸ Ward-Pair Cluster ▸ All Sample ▸ Segment + Year FE

Main Results: Segment-Localized Border FE

Specification: $\ln(Y_i) = \beta \cdot \text{Uncertainty}_i + \alpha_{\text{zone}} + \alpha_{\text{segment}} + \alpha_{\text{year}} + \varepsilon_i$

Sample: Within 500 ft, multifamily only. Main SEs clustered at boundary segment level.

	ln(FAR)	ln(DUPAC)
	(1)	(2)
Uncertainty Index	-0.10***	-0.07**
	(0.04)	(0.04)
Observations	741	741
Dep. Var. Mean	2.48	81.90
Ward Pairs	98	98
Zoning Code FE	✓	✓
Segment FE	✓	✓
Year FE	✓	✓

Interpretation: A 1 SD ↑ in uncertainty is associated with lower multifamily FAR and DUPAC.

From Density to Rents

So far: More uncertain aldermen $\rightarrow$ lower multifamily density

Next question: Does this supply constraint feed into higher housing costs?

Approach: Apply the same border-pair FE design to rental listings

- Data: RentHub rental listings, 2014-2022
- 2.67 million listing observations, 141 ward pairs, BW = 500 ft
- Ward-pair $\times$ year-month FE; SEs clustered at ward-pair level

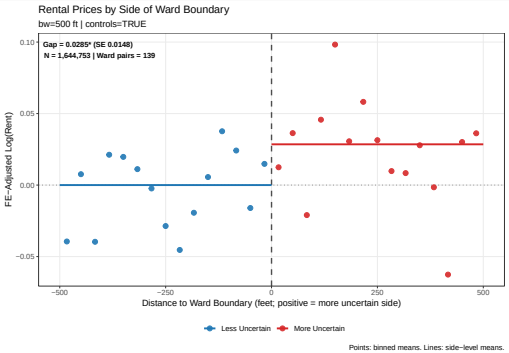
Effect on Rental Prices

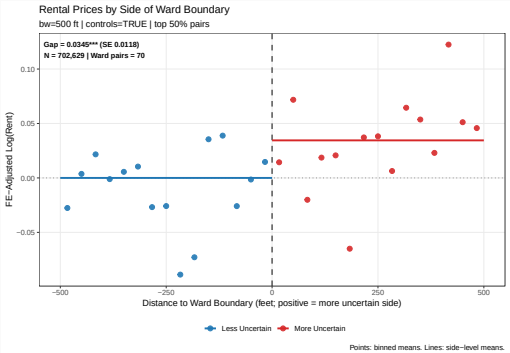
	(1)	(2)
Uncertainty Index	0.028***	0.020**
	(0.010)	(0.009)
Observations	2,664,657	1,644,753
Dep. Var. Mean	2521	2799
Ward Pairs	140	139
Hedonic Controls		✓
Ward-Pair × Year-Month FE	✓	✓

⇒ A 1 SD ↑ in uncertainty ⇒ 3.2% higher rents (no controls) or 1.5% (with controls).

▸ Permutation Test

Spatial RD: Rental Prices at Ward Borders





$\Rightarrow$ Discrete rent jump, gets more pronounced if restricted to above-median uncertainty pairs.

Home Sale Prices: Precise Null

	(1)	(2)
Uncertainty Index	0.005	0.004
	(0.005)	(0.005)
Observations	100,898	100,796
Dep. Var. Mean	387998	387717
Ward Pairs	158	158
Hedonic Controls		✓
Ward-Pair × Year-Quarter FE	✓	✓

⇒ Same design applied to home sales → precisely estimated null.

Event Study

Ruling Out Quality Sorting

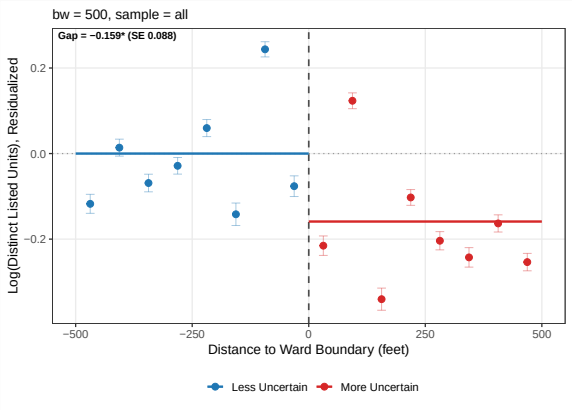
Concern: Rent premium could reflect *better* units on the more uncertain side

Test: Same border-pair FE on unit characteristics (BW = 500 ft, 2014–2022)

	Sqft	Beds	Baths	Multi-Family	Laundry	Gym
More Uncertain Side	-13.5540	-0.0163	-0.0198	0.0218**	0.0254*	0.0180
	(29.0564)	(0.0474)	(0.0437)	(0.0109)	(0.0129)	(0.0219)
Observations	2,042,193	2,674,081	2,669,747	2,674,081	2,674,081	2,674,081
Dep. Var. Mean	894.63	1.37	1.30	0.76	0.59	0.46
Ward-Pair × Year-Month FE	✓	✓	✓	✓	✓	✓

⇒ Core size/bed/bath margins are close to zero; composition shifts are small.
Quality sorting is unlikely to explain the full rent premium at ward boundaries.

Supply Channel: Fewer Available Rentals



	PPML: Distinct Units
Uncertainty Index	-0.1011*
	(0.0573)
Obs.	18,284
Hedonic Controls	✓
Pair × Month FE	✓

Outcome: Distinct listed units (unit-proxy key), ward-pair × side × month
Interpretation: 1 SD ↑ in uncertainty ⇒ ∼12% fewer listed units

Discussion & Policy Implications

Core problem: Regulatory uncertainty and permitting frictions raise the cost of housing development

- Longer permitting times $\rightarrow$ higher carrying costs for developers
- Fewer projects initiated $\rightarrow$ constrained rental supply $\rightarrow$ higher rents
- Effects concentrated in the rental market, not capitalized into sale prices

Policy alternatives to reduce regulatory friction:

1. **Streamline permitting**: Reduce discretion and processing times to lower uncertainty for developers
2. **Transfer decision-making authority**: Shift zoning decisions to the citywide level (i.e. mayor, zoning board)
3. **Citywide preemption**: Override local vetoes for certain housing types (e.g., CA SB-35, MA Ch. 40B, federal H.O.M.E. Act proposals)

Conclusion

Summary of findings:

1. Aldermen vary substantially in regulatory uncertainty (permit processing times)
2. More uncertain aldermen $\Rightarrow$ lower multifamily FAR and DUPAC at borders (about 7–10% in border FE)
3. $\sim$1.5-3% higher rents which appears to be driven by supply

Implications:

- Regulatory uncertainty in decentralized zoning lowers supply and raises rents
- Within-city fragmentation matters, not just across municipalities
- Reforms should target permitting frictions and reduce aldermanic discretion

Future work:

- Model developer location choice decision to run counterfactuals on policy reforms

Thank you!

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Permit Processing Time Residualization

	Log Processing Time (1)
Median HH Income ($10k)	-0.001
	(0.007)
Share Black	0.049
	(0.076)
Share Hispanic	0.108**
	(0.054)
Share White	0.414***
	(0.084)
Homeownership Rate	0.130*
	(0.070)
Share Bachelor's+	-0.186***
	(0.070)
Population (10k)	-0.036***
	(0.013)
Dist. to CBD (km)	-0.013***
	(0.002)
Dist. to Lake (km)	0.0008
	(0.002)
CTA Stations (800m)	-0.019***
	(0.003)
n_permits_wm_l1	-0.001***
	(0.0004)
Observations	139,493
R^2	0.10968
Review Type FE	✓
Permit Type FE	✓
Month FE	✓

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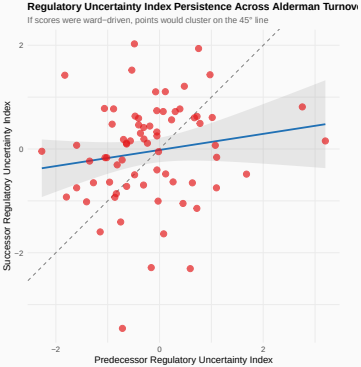
Validation Checks: Permits and Within-Ward Correlation

Outcome (Ward-Month)	Coef. on Uncertainty
N permits (all high-discretion)	-10.29^{***}
	(0.42)
Share with any unit change	0.0059^{***}
	(0.0009)
Units reduced per permit (text)	0.0061^{**}
	(0.0029)
Ward-month observations	10,746–10,753



Predecessor-successor correlation: $\rho = 0.27$

Within-ward SD is 0.72 of overall SD.

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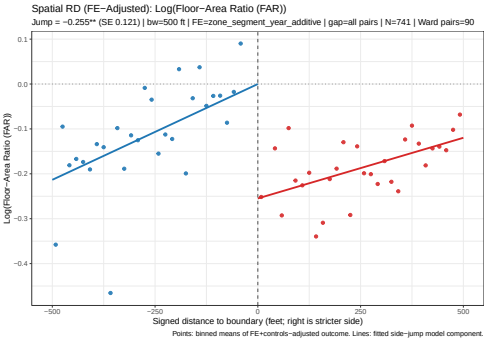
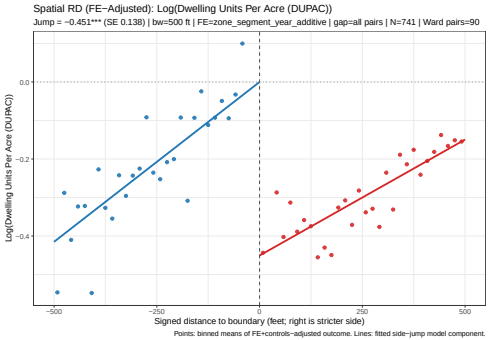
FE-Adjusted Spatial RD: Ward-Pair Cluster

Log(FAR)



Log(DUPAC)



Same multifamily + zoning + segment + year FE; SEs clustered by ward pair.

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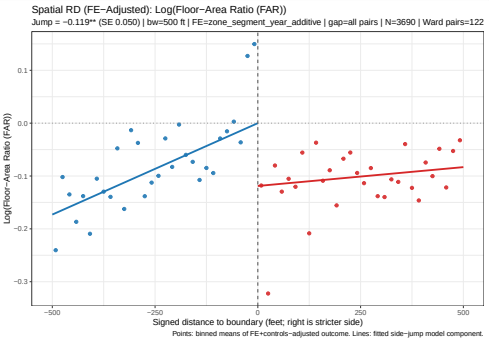
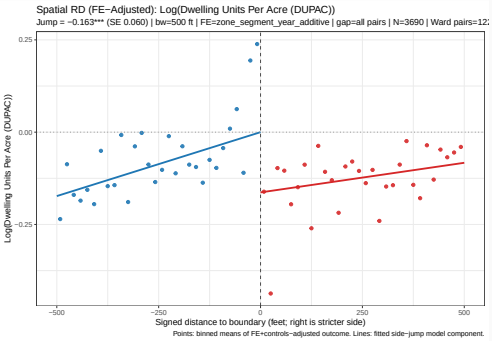
FE-Adjusted Spatial RD: All-Sample Version

Log(FAR)



Log(DUPAC)



Same FE/cluster stack as main slide, but using all parcels with unitscount > 0.

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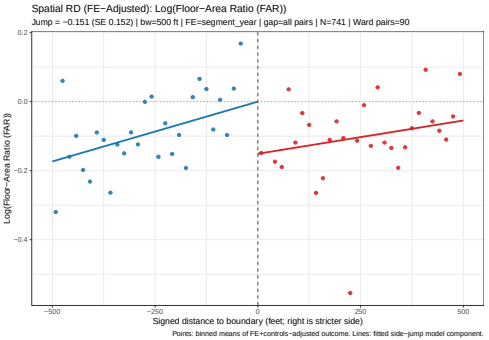
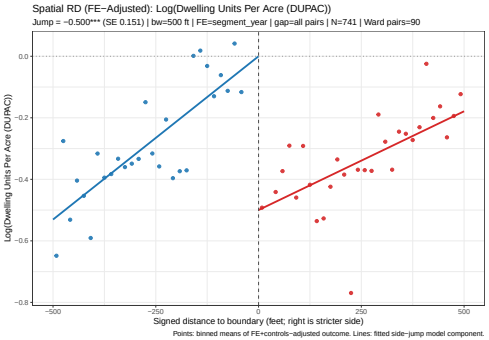
FE-Adjusted Spatial RD: Segment + Year FE

Log(FAR)



Log(DUPAC)



Removes zoning FE; signs remain negative.

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Border FE Robustness: Ward-Pair Cluster

	ln(FAR)	ln(DUPAC)
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Year FE	✓	✓

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Border FE Robustness: Segment + Year FE

	ln(FAR)	ln(DUPAC)
	(1)	(2)
Uncertainty Index	-0.10***	-0.09**
	(0.04)	(0.04)
Observations	741	741
Dep. Var. Mean	2.48	81.90
Ward Pairs	98	98
Segment FE	✓	✓
Year FE	✓	✓

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Border FE Robustness: All-Sample FE Stack

	ln(FAR) (1)	ln(DUPAC) (2)
Uncertainty Index	-0.01	0.0005
	(0.02)	(0.02)
Observations	3,690	3,690
Dep. Var. Mean	1.28	32.01
Ward Pairs	134	134
Zoning Code FE	✓	✓
Segment FE	✓	✓
Year FE	✓	✓

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Border FE Robustness: All-Sample + Ward-Pair Cluster

	ln(FAR)	ln(DUPAC)
	(1)	(2)
Uncertainty Index	-0.01	0.0005
	(0.01)	(0.02)
Observations	3,690	3,690
Dep. Var. Mean	1.28	32.01
Ward Pairs	134	134
Zoning Code FE	✓	✓
Segment FE	✓	✓
Year FE	✓	✓

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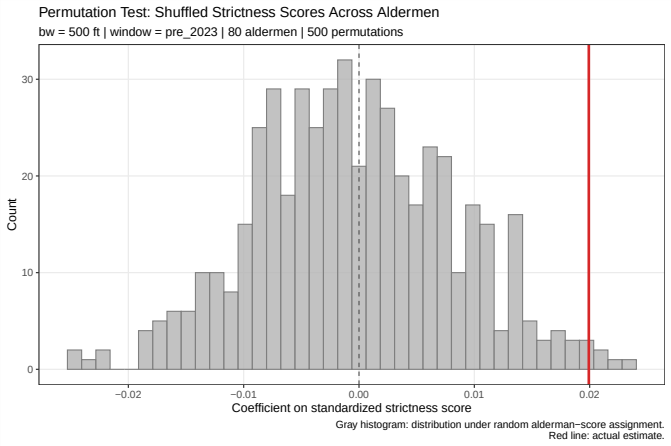
Permutation Test: Rental FE (Controls)



2014–2022 sample, 500 ft bandwidth, controls included.

Permutation p-values: two-sided = 0.02; one-sided = 0.01.

Treatment and Control Block Details

	To Stricter	To Lenient	Total Treated	Control
2015 Cohort				
N Blocks	5,696	5,973	11,669	24,342
Controls per Treated		—		2.09
Mean ΔStrictness	$+0.66$	-0.58	$+0.02$	0.00
2023 Cohort				
N Blocks	3,721	4,480	8,201	23,446
Controls per Treated		—		2.86
Mean ΔStrictness	$+0.87$	-0.72	$+0.00$	0.00

Treatment defined as blocks that switched aldermen during redistricting.
Control blocks are near ward boundaries but did not switch.

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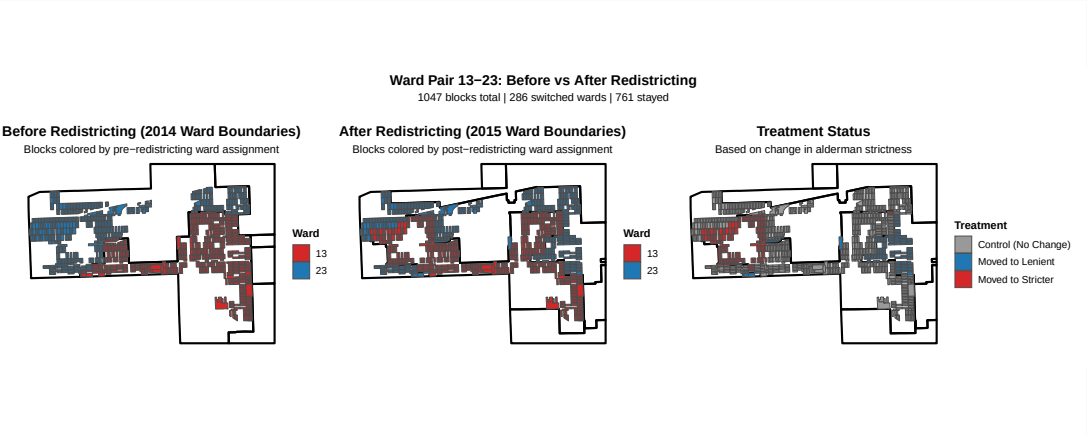
Identification Example: Ward Pair 13–23



Same border pair, before/after redistricting, highlighting treated and control blocks.

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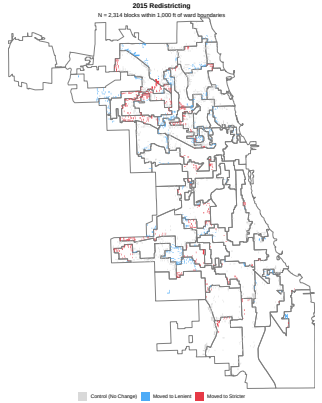
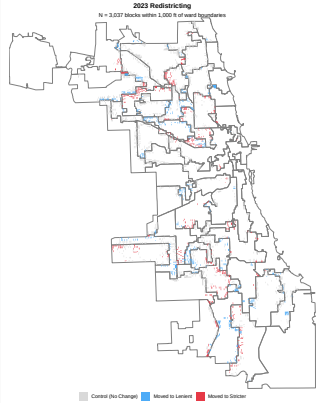
Citywide Treatment and Control Blocks

2015 Redistricting



2023 Redistricting



Red = switched to stricter; Blue = switched to lenient; Gray = control (no switch)

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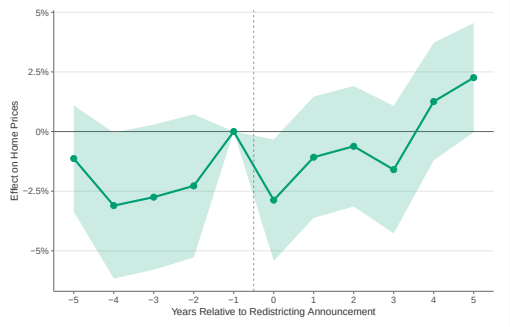
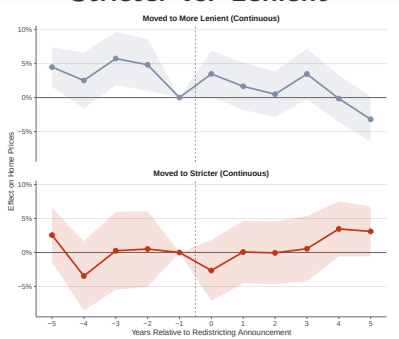
Event Study

Home Price Event Study: Without Hedonic Controls

Aggregate Treatment Effect



Stricter vs. Lenient



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DiD Table

Difference-in-Differences: Home Prices (1000ft, Uniform)

	No Hedonics	Hedonics
Post × ΔStrict	0.003	0.004
	(0.006)	(0.005)
N	78,347	78,317
Dep. Var.	Mean: $339K / Median: $268K	

Stacked announcement specification, 1000 ft bandwidth, uniform weighting.

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